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Antimicrobial resistance in urban river ecosystems

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ABSTRACT

Antimicrobial resistance (AMR) with the ability to thwart clinical therapies and escalate mortality rates is emerging as one of the most pressing global health and environmental concerns. Urban rivers as an important subsystem of the environment offer galore of ecological services which benefit the city dwellers. However, with increased urbanization, industrialization, and heavy discharge of anthropogenic waste harboring antibiotics, heavy metals, pesticides, antibiotic resistance genes (ARGs), antibiotic resistant bacteria (ARB), urban rivers are becoming major reservoirs of ARGs and a hotspot for accelerated selection of ARB. These ARGs in urban rivers have the potential of being transferred to clinically important pathogens. In addition, urban rivers also act as important vectors for AMR spread. This is mainly due to the direct exposure of humans and animals to the heavily contaminated river water and high mobility of organisms (aquatic animals, pathogenic, non-pathogenic bacteria) as well as the genetic elements including ARGs and mobile genetic elements (MGEs) in the river. However, in spite of recent advocacy for comprehensive research programs aimed to investigate the occurrence, extent and major drivers of AMR in urban rivers globally, such studies are missing largely. This review encompasses the issues of AMR, major drivers and their vital roles in the evolution and spread of ARB with an emphasis on sources and hotspots of diverse ARGs in urban rivers contributing to co-occurrence of ARGs and MGEs. Further, the causal factors leading to adverse effects of antibiotic-load to river organisms with an elaboration on the current measures to eradicate the ARB, ARGs, and remove antibiotics from the urban river ecosystems are also discussed. A perspective review of current and emerging strategies with potentials of combating AMR in urban river ecosystems including advanced water treatment methodologies and floating islands or constructed wetlands.

1. Introduction

Antimicrobial resistance (AMR), a phenomenon where microorganisms no longer respond to the available antimicrobial drugs, has increased dramatically in recent years and is considered as one of the greatest global health threats of the 21st century (Yu et al., 2019). According to the Centers for Disease Control and Prevention (CDC) more than 2.8 million AMR infections occur and more than 35,000 people die due to the infections caused by resistant microbes each year in USA (Frieden, 2019). Globally, approximately 700,000 deaths are attributed annually to AMR and are projected to reach 10 million by 2050 if left unchecked leading to global financial burden to nearly US$100 trillion (Li et al., 2021). Murray et al. (2022) assessed the global burden associated with AMR infections across and estimated 4.95 million deaths, of which 1.27 million deaths were directly attributable to drug resistance. In other words, if all AMR infections were replaced by no infection, 4.95 million deaths could have been prevented in 2019, whereas if all drug-resistant infection were replaced by drug susceptible infection, 1.27 million deaths could have been prevented.

Traditional views of AMR, antibiotics produced and administered into the humans, companion animals, livestock, crops, plants, and commercial and laboratory settings serves as a source to pollute the environment and agriculture resulting in the development of drug-resistant microbes and infecting the systems of humans and other

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organisms (Manyi-Loh et al., 2018; Larsson and Flach, 2022). However, the classic concept of AMR incorporates the consequences and implications of their possible species interactions such as competition, mutualism, predation, parasitism, and complex associations as an ecological as well as evolutionary aspect (Banerji et al., 2019). Studies on interactions between antibiotic-resistant and susceptible-strains in the pathogenic *Pseudomonas aeruginosa* bacteria for the selection of intracellular resistance to streptomycin and extracellular resistance to carbenicillin resulted in the corporation and competition of the strong and weak bacterial colonies for the survival of the fittest (Frost et al., 2018; Bottery et al., 2021). The corporative and mutual behavior between the two strains of *Escherichia coli* (resistant and sensitive) forming an effective cross-protection mutualism by allowing resistant cells to protect sensitive cells from antibiotics suggest their equal importance clinically as well ecologically (Yurtsev et al., 2016; Li et al., 2019; Adamowicz et al., 2020). Similarly, predation and parasitism (Leisner et al., 2016), and complex associations (Banerji et al., 2019) of antimicrobial-resistant bacterial communities have been effective under certain circumstances in natural environments.

Urban rivers considered as the cradle of city civilization, power source of water, and the pollution purification place, have emerged recently major drivers of AMR prevalence (Xu et al., 2016; Zhou et al., 2017). With rapid industrialization and urbanization, the urban rivers have been greatly influenced by anthropogenic activities via the massive discharge of industrial and domestic wastes containing antibiotics, heavy metals, biocides, pesticides leading to the growth of antibiotic resistant bacteria (ARB) with the strong pool of antibiotic resistance genes (ARGs) showing their adverse effect on human-health and environment (Jia et al., 2018; Zhang et al., 2020b).

Chen et al., (2019a,2019b) detected a total of 203 ARGs subtypes with the dominance of multidrug, macrolide-lincosamide-streptogramin, quinolone, bacitracin, and sulfonamide resistance genes in urban river sediments with various metal/ biocide resistance genes, mobile genetic elements (MGEs) including transposons, integrons, and plasmids. However, recently strong associations of ARGs with metal/ biocide resistance genes and MGEs have also been detected by the co-occurrence analysis using ARG-carrying contigs (ACC) indicating active horizontal gene transfer (HGT) of ARGs in urban river ecosystem (Chen et al., 2019a). Further, Jia et al. (2018) detected high frequencies/ concentration of antibiotics/ pharmaceuticals/ drugs including quinolones, macrolide, sulfonamide, and tetracycline coupled with predominance of ARGs viz. *tetA, tetC, tetZ, sul1, gyrA, cmlA, blaTEM*, and *ermF* in an urban river of Xi'an, China. Frequent release of anthropogenic and industrial waste into the urban rivers has left severe impacts on the aquatic ecosystems and human health. For instance, Zhou et al. (2022) investigated the impacts of urbanization on ARG dissemination in Jiulong River basin (China) via studying the ARG-profiling and AMR-patterning. From 21 sampling sites, a total of 244 ARGs and 12 MGEs were detected. AMR abundance was found to be significantly increased at sampling points near urban conurbations. However, overall resistome abundance decreased as the river flowed downstream. The downstream reduction of the resistome was most likely due to the dilution effects through mixing river water with several river confluences. Further, antibiotic resistome abundance was positively correlated with city size, and fast expectation maximization microbial source tracking, suggesting that the majority of the antibiotic resistome originated from anthropogenic activities (Zhou et al., 2022). The anthropogenic activities, such as urbanization and associated contamination, was projected as the main driver of environmental dissemination of ARGs, and thus the AMR. Almakki et al. (2019) reviewed the environmental pressures exerted on bacterial communities in urban runoff waters while discussing the impact of these settings on AMR. The authors urged to identify the environmental reservoirs of AMR and ARGs to deepen our understandings of the epidemiological cycle and of the dynamics of urban AMR.

Bioaccumulation of pharmaceuticals such as antibiotics, other drugs, and their metabolites in the liver, gills, and muscles of multiple fishes from urban rivers has been observed (Ondarza et al., 2019) with an abundance of erythromycin (0.7–5.6 μg/kg), sulfamethoxazole, sulfathiazole, trimethoprim (0.9–5.5 μg/kg), norfluoxetine primary metabolite of the antidepressant fluoxetine (1.1–9.1 μg/kg), and caffeine (13 μg/kg). It adversely affects the immune system, microbial flora, growth, reproduction, digestion of aquatic animals residing inside urban rivers leading to increased occurrence, dissemination as well as an escalation of AMR in the environment and eventually in human body (Zheng et al., 2021). Fig. 1 depicts major pathways/resources of antimicrobial resistance emergence and spread in urban river systems.

Urban rivers are turning into frequent sources of AMR spread in humans, livestock, aquaculture, and the environment. This situation demands to have a scientific review on the current scenario of urban river ecosystem concerning AMR, ARGs, and HGTs. Current review focuses on the factors favoring emergence and prevalence of AMR in the urban river ecosystems with multiple sources of ARGs, besides their occurrence and dissemination in the environment. The impacts of antibiotic load, increasing AMR with the co-occurrence of ARGs and MGEs powered by HGTs are also discussed. This review thus provides the reader a perspective review on current condition of AMR, recent developments and potent measures to mitigate it from the urban river ecosystems.

2. Factors favoring emergence and prevalence of antimicrobial resistance

Antibiotics are often discharged in the environment without effective treatment which drives the emergence, prevalence and spread of AMR. Several recent studies have proved the selection of AMR at high and low antibiotic concentrations defined as minimal inhibitory concentration (MIC), the minimal concentration of antibiotics required to inhibit the bacterial growth (Gullberg et al., 2011; Liu et al., 2011; Andersson and Hughes, 2014; Li et al., 2017; Almakki et al., 2019). Sub-MIC, a concentration less than MIC, the selection at which is even more worrisome as it creates a combination of highly resistant and fit mutants that are prone to spread in the absence of antibiotics and thereby easily becoming stable in the population (Sandegren, 2019). Wistrand-Yuen et al. (2018) in a study exposed *Salmonella enterica* to a sub-MIC level of streptomycin leading the bacteria to acquire a high level of resistance. At sub-MIC selection, resistance is induced/ generated via several small effects that are combined to confer a high level of resistance by mechanisms such as reduction of aminoglycoside uptake, alteration of ribosomal RNA target, and induction of the aminoglycoside modifying enzymes (Wistrand-Yuen et al., 2018) indicating the ability of weak selection pressure to cause the evolution of a high level of resistance. In an attempt to assess whether antibiotic (tetracycline) pollution in aquatic environments promote AMR development, Lundström et al. (2016) determined the minimal selective concentrations (MSCs, lowest concentration of antimicrobials that gives the resistant strains a competitive advantage based on growth rates) in biofilms of aquatic bacterial communities. The authors concluded that the selection pressure induced by the antibiotics released into the environment could enrich ARGs and AMR phenotypes and thus increasing the risk for transmission to humans and animals (Lundström et al., 2016). Similarly, another study showed a relative abundance of *sul1* and *sul2* genes positively correlated with total sulfonamide-concentration demonstrating that the antibiotic exerted selective pressure on *sul1* and *sul2* ARGs (Na et al., 2014). Likewise, Xiong et al. (2015) observed positive selection imposed by antibiotics such as sulfonamides, tetracyclines, and fluoroquinolones on tetracycline resistance genes (*tet*M, *tet*W, *tet*X, *tet*O, and *tet* Q), sulfonamide resistance genes (*sul1* and *sul2*) and plasmid-mediated quinolone resistance genes (*oqx*A, *oqx*B, and *aac*(6′)–Ib). The results also highlighted a significant selection of specific

Fig. 1. Major pathways/resources of antimicrobial resistance emergence and spread in urban river systems. ( )Co-pollutants (Heavy Metals, Microplastics, Petroleum by-product, Metallic Paints, Pesticides). ( )Antimicrobial Drugs (Antibiotics. Biocides etc.). ( )Human pathogens. ( )Bacteria from animal, soil, plant and other sources. (HM: Heavy Metals; Ab: Antibiotics; MGE: Mobile Genetic Element; ARG: Antibiotic Resistance Genes; MP: Microplastics).

bacterial communities by the antibiotics including taxa-associated opportunistic pathogens (*Gammaproteobacteria, Clostridia* etc.) (Xiong et al., 2015).

Beside antibiotics as contaminants, an increasing number of reports have suggested heavy metals to be capable of increasing the abundance of ARB and ARGs in the urban river ecosystem as they are also known to exert selective pressure (Ji et al., 2012; Seiler and Berendonk, 2012; Gullberg et al., 2011; Gao et al., 2015; Xu et al., 2017). Unfortunately, unlike antibiotics, heavy metals being not easily degraded can provide long-term selection pressure on environmental micro-organisms contributing to increased prevalence of ARB and ARGs (Xu et al., 2017). Wang et al. (2021) in their study detected an increased abundance of pathogens resistant to ampicillin, downstream of Xiangjiang river influenced heavily by metal mining activities. Similarly, Xu et al. (2017) observed that the heavy metals (like Cu^{2+} and Zn^{2+}), promoted the prevalence of multi-drug resistant (MDR) pathogens in the river waters. Further, Silva et al. (2021) reported an increased prevalence of cefotaxime, tetracycline, and kanamycin resistant bacteria after exposure to copper and zinc, respectively. Zhang et al. (2020a) investigated the shift of antibiotic resistome (comprising all of ARGs) caused by the heavy metal pollution incidents via simulating an arsenic shock loading along with the associated risks imposed on drinking water systems. The resistance achieved without any genetic alteration is known as phenotypic resistance (Corona and Martinez 2013).

Song et al. (2017) compared the ability of tetracycline and metals to select tetracycline resistance in the soil bacterial communities. The authors detected an increased tetracycline resistance in soil containing an environmentally relevant level of Cu (≥365 mg/kg) and Zn (≥264 mg/kg) but not in the soil which was spiked with unrealistically high levels of tetracycline (up to 100 mg/kg), hence demonstrating stronger selective pressure exerted by heavy metals for environmental selection of resistance to antibiotics in soil. It may pose a threat of contaminating natural waterways with heavy metals through runoff and thus may contribute to AMR in water bodies/rivers.

Other factors that promote AMR include disinfection byproducts (DBPs) like pesticides, paints, detergents, and microplastics. Li et al. (2016) analyzed the contribution of common DBPs (chlorite and iodo-acetic acid) in AMR-spread and reported antibiotic like effect of DBPs that led to the evolution of resistant *E. coli* strains under both low (sub-MIC) and high (near-MIC) exposure concentrations. In another study, Berman (2012) investigated the ability of heavy metal-based antifouling paints to co-select ARB, and observed a co-resistance of *Gammaproteobacteria* (a class of bacteria including *Enterobacteriaceae*, *Pseudomonadaceae*, *Vibrionaceae* families etc.) to both heavy metal (copper and zinc) and antibiotic (tetracycline).

Further, Flach et al. (2017) observed enrichment and increased abundance of integron-associated integrase, insertion sequence common regions (ISCR) transposase (elements that are closely related to an unusual family of insertion sequence named IS*91*. This moves by the process called rolling-circle replication) and chromosomal resistance-nodulation-cell division (RND) pump (that efflux out distinct class of antibiotics and provide innate drug-resistance) genes showing strong selection pressure exerted by heavy metal-based antifouling paints and their ability to co-select certain ARB likely by favoring species and providing cross-resistance (Flach et al., 2017). Furthermore, selective enrichment of ARGs (*tet*A, *tet*C, *tet*X, *emr*E, *sul1*, *sul*A/FolP-01) and antibiotics (sulfamerazine, chloramphenicol, tetracycline, and tylosin) were detected on microplastics by Wang et al. (2020d). Similarly, Zhang et al. (2020c) reported 100–5000 times higher ARB count and higher ratio of ARB to total bacteria on microplastic than in water. Therefore, microplastics are emerging as hubs/vectors for AMR spread as they provide ideal niche for biofilm formation, selective enrichment, and increased HGT of ARGs (Kaur et al., 2021). Thus concluding that an elevated level of anthropogenic waste such as antibiotics, heavy metals, pesticides can exert the selection pressure while microplastics provide ideal niche that facilitates the emergence, maintenance, and dissemination of AMR in the environment.

3. Sources of antibiotic resistance genes spread in urban river ecosystem

ARGs have grabbed considerable attention in recent times because of

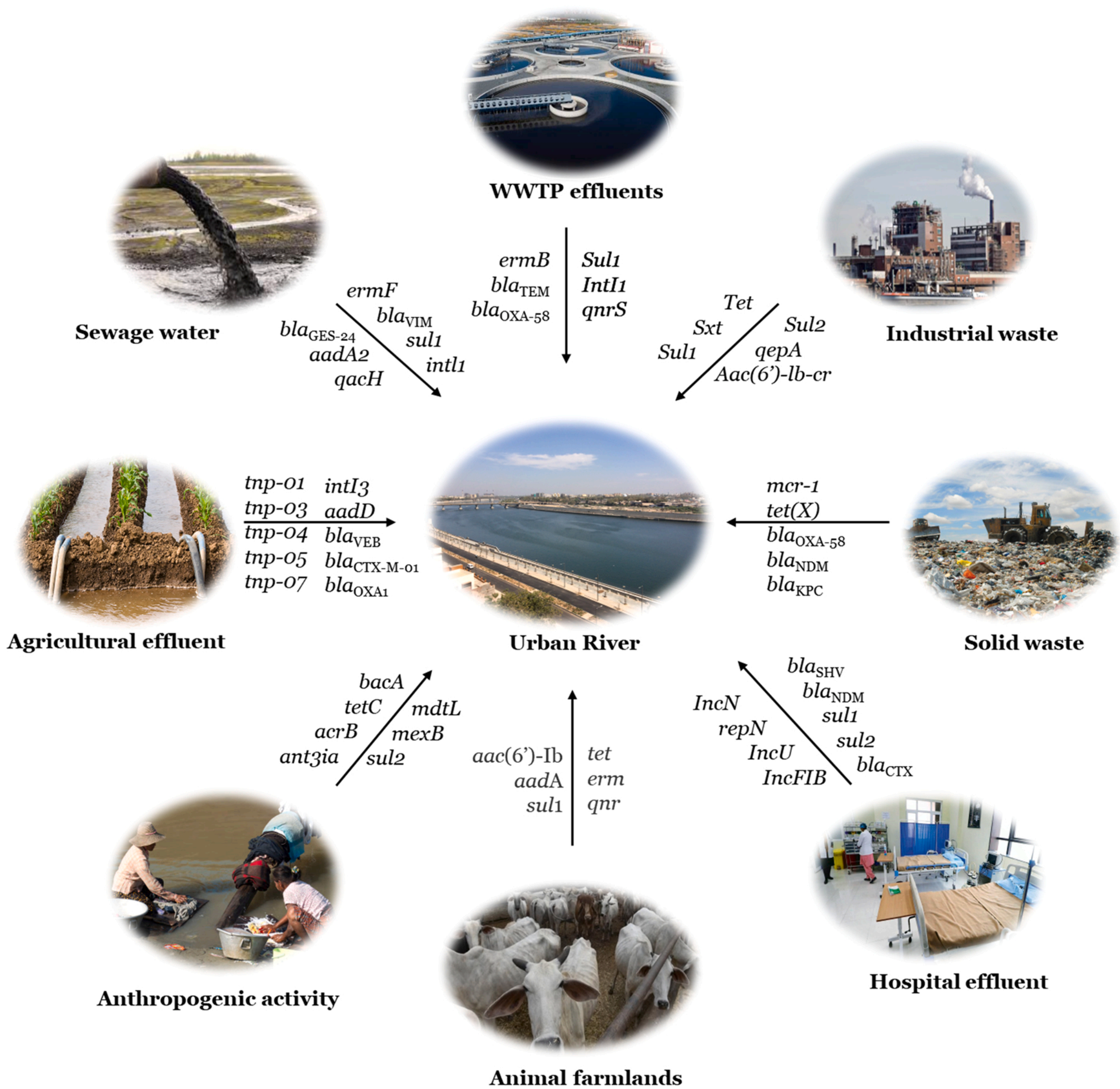

Fig. 2. Various ARGs and MGEs eluted from various sources to the urban river (Zhou et al., 2017; Osińska et al., 2020).

their ability to remain persistent in the environment even if the bacterial cell carrying ARGs are killed. In many cases ARGs are maintained in the microbial population even if the antibiotic selection pressure is removed (Pruden et al., 2006; Zhang et al., 2018; Deng et al., 2020). Further, ARGs can also be exchanged between microbes of clinical and environmental/community origin via HGT (Fang et al., 2019). This prompt the bacteria to develop resistance against all classes of antibiotics and all levels/lines of defenses including last-resort antibiotics.

The urban river ecosystem provides an ideal setting for the emergence, widespread occurrence and spread of diverse ARGs due to frequent pollution by antimicrobial compounds introduced via various factors including health care facilities, livestock farming, landfills, aquaculture (Rodriguez-Mozaz et al., 2015; Zainab et al., 2020) and urban wastewater treatment plant (UWWTP) representing one of the major reservoirs and becoming critical source of ARGs spread into the environment (Fig. 2) (Rizzo et al., 2013). UWWTP receives antibiotics, heavy metals, pesticides, from different sources which can add selective pressure promoting the spread of ARGs. Additionally, high microbial density in UWWTP further facilities ARGs transfer through HGT thus acting as hotspots for the prevalence and dissemination of ARGs (Pérez-Cobas et al., 2016; Cacace et al., 2019; Liu et al., 2019b; Lin et al., 2021). Some leads show that although WTPs can reduce ARB and ARGs, their large part is still transmitted into the receiving water bodies such as rivers, lakes, ponds, etc. (Marti et al., 2013; Berglund et al., 2015; Walia et al., 2016; Osińska et al., 2020; Pantanella et al., 2020). Sabri et al. (2020) investigated the occurrence of antibiotics (tetracyclines, macrolides, sulfonamides), ARGs (*sul1, sul2, tetW, ermB*) and class 1 integrons in Dutch river receiving effluents from WWTP and observed a significant increase in levels of antibiotics and ARGs in the river compared to the upstream water samples. The results showed the persistence of ARGs in water and sediments till 20 km from WWTP effluent discharge point (Sabri et al., 2020). Similarly, Osińska et al. (2020) compared the prevalence of antibiotics, ARB, and ARGs in river water/ wastewater and detected dominance of bla_{TEM}, *tetA, sul1* genes in the river water after the inflow of WWTP effluent.

Another hotspot for ARGs seem to be the hospital effluent/wastewater. Rodriguez-Mozaz et al. (2015) investigated the pollution level of a broad range of antibiotics and ARGs from the hospital and urban wastewaters, their removal through WWTP, and their presence in receiving river. Authors reported the highest concentration of fluoroquinolones especially in hospital effluent samples and bla_{TEM}, *qnrS, sul1, ermB, tetW* ARGs in both hospital effluents as well as WWTP influent samples (Rodriguez-Mozaz et al., 2015). The incomplete removal of antibiotics and ARGs in WWTP severally affected the receiving river, where both types of emerging pollutants were detected at higher concentrations in downstream waters than in samples collected upstream from the discharge point (Rodriguez-Mozaz et al., 2015). The results demonstrate the widespread occurrence of antibiotics and ARGs in urban and hospital wastewater and how these effluents/wastewater even after treatment contribute to the spread of antibiotics and ARGs in river ecosystems.

Animal husbandry represents additional subscribers of ARGs in the environment with a major portion being shared by poultry wastewater. Zhao et al. (2020) analyzed fresh feces from common birds in urban environment as well as human community and reported an abundance of *tet*W, bla_{TEM}, *sul*1, and *intI*1 (up to 10^9, 10^8, 10^9, and 10^{10} copies/g dry feces) with relative concentration comparable to livestock and poultry bird that are occasionally fed antibiotics, highlighting the contribution of wild urban bird feces for the spread of ARGs.

Constructed wetlands designed to treat wastewater recently has come up as an important hub or carrier of ARGs. For instance, Song et al. (2018) assessed the removal of sulfamethoxazole, tetracycline and the fate of ARGs using vertical up-flow constructed wetland, which exhibited high removal efficiency for the antibiotics. Results also indicated the increase in the abundance of ARGs (mainly *sul*- and *tet*- related genes) with increased operation time. Further, a positive correlation was seen with the abundance of ARGs and accumulation of antibiotics in different layers of vertical up-flow constructed wetland with highest ARGs (*sul* and *tet*) being at the bottom layer due apparently to the exposure of high antibiotics concentration (Song et al., 2018).

Application of biosolids, manure, wastewater (for irrigation) contaminated with antibiotics, ARB harboring ARGs, and irrational use of antimicrobials in agricultural soil creates sustained selection pressure that facilitates the emergence and prevalence of ARGs in soils (Osbiston et al., 2021), that eventually gets transmitted into the rivers through rainfall and runoff making agricultural soil as a principal source for the dissemination of ARGs into the river. Studies have reported an increased abundance of ARGs (*aad*A1, *aad*A2, *mph*A, *sul1, sul2*, *tet*G) in agricultural soils impacted with anthropogenic activity such as wastewater irrigation, manure, biosolid utilization (Pan and Chu, 2018; Shi et al., 2020; Sun et al., 2020; Wang et al., 2020a; 2020c).

Additional sources of ARGs include urban watersheds, streams, ponds, lakes, and other water bodies that help in the spread and enrichment of ARGs. Hou et al. (2021) in a study examined the abundance and diversity of ARGs in urban water ecosystems that include drinking water reservoirs, landscape pond, influent and effluents of WWTP from the coastal city identifying total of 237 ARGs with an abundance of multidrug, beta-lactamase, and aminoglycoside resistant genes. Among examined urban water habitats, influents from WWTP have been examined to be with the great abundance of ARGs (Yin et al., 2019), followed by urban ponds, effluents, and then drinking water reservoirs (Ma et al., 2017). Additionally, urban ponds exhibited more exogenous pollution caused by anthropogenic activity and the strongest co-occurrence relationship between ARGs and MGEs indicating urban ponds as a hotspot for ARGs dissemination (Hou et al., 2021). Likewise, Yang et al. (2017) analyzed the distribution of *sul*, *tet*, quinolone (*qnr*) resistant genes, their relationship with heavy metals (Cr, Ni, Cu, Zn, As, Cd, Hg, Pb) and ARGs the authors detected *sul*1*, sul*2 (dominant ARGs accounted for 86.28–97.79% of total ARGs abundance), *tet*A*, tet*B*, tet*C*, tet*G, and *qnr*D in all six lakes highlighting co-selection of heavy metals as a major factor driving the propagation of ARGs in urban lake ecosystems.

4. Impact of increased antimicrobial resistance and antibiotic load

Antibiotic use, particularly in healthcare, veterinary and agriculture can result in the release of antibiotics into surface waters, promoting the development of AMR in the environment, and potentially leading to human, animal and aquatic health risk. The increased frequency of ARB and resistant phenotypes in the environment, often termed as “antibiotic resistance pollution” limits the ability of present antibiotics to fight against infectious diseases. ‘Superbugs’ are known for carrying resistance genes against multiple classes of antibiotics (Davies, 1996) that are often associated with the presence of integrons and MGEs. One of the major factors that increase the antibiotic load and its resistance in the environment is injudicious usage of antibiotics in veterinary, agriculture and allied industries making it difficult to track the antibiotic usage in general populace and in hospitals. Sewage from both hospitals and general populace are combined and transported to WWTP thus acting as hotspot for AMR transmission. Y. Chen et al. (2020); H. Chen et al. (2020) reported occurrence of antibiotics, human pathogens and their antibiotic resistance determinants in waste water treatment plants of Nanjing, the Jinchuan River and the Yangtze River, China.

Addressing the rising threat of AMR requires a holistic and multisectoral approach, referred to as the “One Health” concept considering the health of humans, animals and environment (Robinson et al., 2016). Studies have reported the presence of antibiotics in the chlorinated water like macrolides and quinolones that accumulates within human populations via long-term exposure to drinking water, food, or consumer goods leading to health issues like nausea, diarrhea, etc. (Ye et al., 2007). Antimicrobial compound such as triclosan used in cloths and soaps, has been increasingly detected in stream and rivers around the world (Bever et al., 2018).

5. Co-occurrence of antibiotic resistance genes and mobile genetic elements in urban rivers

Co-occurrence of genes is the phenomenon of expression of genes together, for example if ARG and metal/biocides gene physically located on same plasmid, metal/biocide exposure can promote HGT of ARG. With the increased establishment and dissemination of antimicrobial resistant microorganism in the urban river ecosystems, it has become a growing concern. It is widely known that all pathogenic, commensal, and environmental bacteria carrying specific MGEs and bacteriophages form a reservoir of ARGs from which bacterial pathogens can gain resistance via HGT (Von Wintersdorff et al., 2016). Studies on high throughput screening of ARGs and MGEs collected from urban rivers of Japan have revealed their co-occurrence patterns. The transmission of ARGs in pathogenic bacteria has been widely aided by a variety of HGT phenomena including conjugation, transduction and transformation developing AMR capabilities in them (Vrancianu et al., 2020). River being major reservoirs of ARGs are important for disseminating them with MGEs further playing an important role in the AMR development and its emergence. Recent investigations on the relationship between the ARGs and MGEs in the upper reaches of Huaihe river, China resulted in the identification of the *int*I-1(clinic), *IS*613, *Tp*614, *tnp*A.01, and *tnp*A.04 MGEs as the major drivers of the ARGs revealing their likelihood in transferring ARGs (*sul*1, *sul*2, *tet*M-01, *erm*B, *str*A, *str*B, *tet*G and bla_{OXA-10}) amongst the bacterial strains (Zhang et al., 2021). Similarly, co-occurrence of various MGEs and ARGs (*tnpA*-07, *tnpA*-01, and *tnpA*-03) have been reported from rural river of Southeast China (Na et al., 2021), East Tiaoxi River, China (Zheng et al., 2017), the sediments of an interconnected river-lake system (Fuhe River and its receiving Lake Baiyang in northern China) (Y. Chen et al., 2020; H. Chen et al., 2020), peri-urban river (Zheng et al., 2018), and the sediments of Lake Tai, China (Chen et al., 2019b).

Further, Chen et al. (2019c) reported the effect of seasonal variations in the abundance and distribution of genes encoding transposase (*tnp*A-02, *tnp*A-07, *tnp*A-05, *tp*614, and *int*I-1) in drinking water river-reservoir system revealing their contribution in shaping the profile of ARGs. Partridge et al. (2018) reviewed how MGEs including insertion sequences, transposons, and gene cassettes/integrons are able to transfer between bacterial cells via plasmids and integrative conjugative elements. Authors advocated that together these elements play key role in facilitating the horizontal genetic exchange and therefore promote the acquisition and spread of ARGs (Partridge et al., 2018). Similar studies were reported in Taipu River and Jinze Reservoir with abundance of 192 ARGs and 12 MGEs in seasonal variations with MGEs encoding integrons (*intI-1(clinical), intI-1 LC* and *intI-3*) and transposases (*tnpA-01, tnpA-02, tnpA-03, tnpA-05, tnpA-07* and *TP614*) being the most common (Li et al., 2020). Li et al. (2018) investigated the occurrence and distribution of integrase genes and ARGs in sediment samples collected from drinking water sources, urban river and coastal areas of Zhuhai, China, and reported the highest concentration of *sul2* (average concentration of 3.78×10^7 and 9.04×10^7 copies/g of sediment in both dry and wet seasons) followed by *tet*A, *tet*C, *tet*O, *erm*B, *dfr*A1 and bla_{PSE-1}. Some of the examples with the co-occurrence of ARGs and MGEs are given in Table 1.

6. Current measures to mitigate antimicrobial resistance in urban river ecosystems

Wastewater from clinical, industrial, pharmaceutical, animal husbandry, and other sources have been observed with elevated levels of antibiotics, ARGs, drug-resistant bacteria, which eventually find their way into an aquatic ecosystem (including river) accelerating the emergence and spread of antimicrobial resistance. Thus in order to mitigate

Table 1
Antibiotic resistance genes (ARGs) and mobile genetic elements (MGEs) present in urban river.

Urban river	Antibiotic resistance genes (ARGs) and their abundance	Mobile genetic elements (MGEs) and their abundance	Microorganism/Pathogen	References
Ganga, Gomti, Yamuna, and Hindon river, India	*dfrA-dfrB* (37.5%), *sul1* (9.76%), and *sul2I* (2.44%)	*intI1* (89.25%) and *intI2* (1.07%)	*V. parahaemolyticus, E. coli* K-12	(Chaturvedi et al., 2021)
Lake Tai and Lake Baiyang urban lakes, China	Quinolone (36.8 ×/Gb), trimethoprim (40.3 ×/Gb), tetracycline (*tetB, tetB(P)*) (32.3 ×/Gb), sulfonamide (*sul4*) (27.2 ×/Gb), colistin (*mcr-1, mcr-2, mcr-3, mcr-4, mcr-5, mcr-6, mcr-5.1*) (4.2 ×/Gb), and carbapenem resistant genes ($bla_{OXA-198}$) (<4.0 ×/Gb)	*tnpA, IS91, istB* and *intI1* (50%)	*Pseudomonas stutzeri, Ralstonia pickettii, Sutterella wadsworthensis, Pseudonocardia autotrophica, Mycobacterium kansasii,* and *Rhodococcus rhodochrous*	(Chen et al., 2021)
Mula-mutha river, India	*set1A, set1B, sen astA, aggA, aafA, pet, stx1* and *stx*	Class 1, 2 integrons (*intI1, intI2*)	*E. coli*	(Namachivayam et al., 2021)
River Liffey and River Tolka, Dublin, Ireland	bla_{TEM} (1.1×10^4 to 4.0×10^4 GC/100 mL), bla_{SHV} (2 $\log_{10}$ GC/100 mL), *qnrS* (3 $\log_{10}$ GC/100 mL), and *sul1* (3.2×10^4 to 7.6×10^4 GC/100 mL)	–	*E. coli*	(Sala-Comorera et al., 2021)
Sundarijal, Thapathali, and Chovar urban river, Nepal	bla_{TEM} (4.1 ± 2.0 log copies/mL), ermF (3.8 ± 1.7 log copies/mL), tetA (2.5 ± 1.7 log copies/mL)	*intI1* (3.4 ± 1.4 log copies/mL and 5.6 ± 0.6 log copies/mL upstream and midstream respectively)	*E. coli*	(Thakali et al., 2020)
Yamuna river, India	bla_{TEM-1}, $bla_{CTX-M-15}$, bla_{CMY-42}, *dfrA17, dfrA1, dfrA7, aadA5, aadA1, aadA2,* and *aacA4*	*intI1, ISEcp1,* and *IS26*	*E. coli*	(Singh et al., 2018)
Stroubles Creek in Blacksburg, Virginia, USA	*sul1, sul2, tet(O), tet(W), 21* and *erm(F)* (0.0222; 0.0069; 0.0002; 0.0103 and 0.0074 gene copies per L, respectively)	–	*E. coli* and Enterococci	(Garner et al., 2017)
Mutha river, India	Amphenicols (74-fold), macrolides (64-fold), tetracyclines (60-fold), beta-lactams (41-fold), aminoglycosides (21-fold), sulfonamides (22-fold) and trimethoprim resistance genes (9-fold), tigecycline resistance gene tet(X) (>206-fold), carbapenemases of the GES-type (>630-fold) and OXA-58 (>23-fold).	integron-associated integrases (2-fold) and ISCR transposases (6-fold)	*Acinetobacter*	(Marathe et al., 2017)

AMR from the rivers, effective wastewater treatment strategies are a must (Singh et al., 2019). Treatment strategies such as constructed wetland, disinfection by peracetic acid, ozonation, fenton oxidation, and other advanced oxidation processes (AOP) required to eliminate ARB, ARGs, and antibiotics are listed in the Table 2 and further delineated in detail in this section.

6.1. UV disinfection

UV based treatment strategies have been proven to be effective in inactivation of antibiotics, and ARB. Complete degradation of ciprofloxacin (within 15 min under the optimum concentration of oxidant H_2O_2 equal to 10 mg L^{-1}); inactivation of *E.coli* and ciprofloxacin (CIP)-resistant *E. coli* using UV-C/H_2O_2 (within 2 min) (Boudriche et al., 2017); complete degradation of erythromycin (ERY) and total inactivation of ERY-resistant *E. coli* (63%) within 90 min by using highly efficient UV-C-driven advanced oxidation process via sulfate radicals ($SO4^{\bullet-}$) production have been investigated till now (Michael-Kordatou et al., 2015). Similarly, Ferro et al. (2016) reported bacterial inactivation and reduction of ARGs in urban wastewater with the help of UV/H_2O_2 treatment. However, this process was not found to be effective in the removal of ARG like *bla*$_{TEM}$ and poorly affected the copies (4.3×10^4 copies mL^{-1} after 240 min treatment) of *qnrS* gene.

6.2. Peracetic acid disinfection

Peracetic acid (PAA) is an emerging disinfectant as a potent alternative to chlorination disinfectants with an assertion that its spontaneous breakdown produces only harmless disinfection by-product such as acetic acid, water and oxygen (Ölmez and Kretzschmar, 2009). Turolla et al. (2018) reported an effective reduction of antibiotic-resistant total heterotrophic bacteria (THB) by PAA. Similarly, Balachandran et al. (2021) reported a reduction of MDR *E. coli*, *Enterococci*, and no regrowth of bacteria after 72 h from PAA disinfection. Additionally, PAA when compared with chlorine showed lower decay suggesting the formation of less-toxic disinfection byproduct. PAA coupled with photo-driven advanced oxidation process (AOP) has been recently investigated as a tertiary treatment for urban wastewater revealing their low dose effectiveness with UV processing for the elimination of drug-resistant *E. coli* when compared to sunlight in groundwater (Rizzo et al., 2019). Similarly, Zhang et al. (2019) assessed the functionality of bacterial plasmid DNA in vivo after PAA disinfection and compared with chlorination revealing neither PAA nor chlorine at a dose, high enough to inhibit bacterial regrowth, could completely destroy the plasmid DNA harboring ARGs. However, residual plasmid DNA was still observed intact and functional showing AMR and their transforming ability. Similarly, several studies confirmed no effect of PAA disinfection on the relative abundance of ARGs (Di Cesare et al., 2016; Luprano et al., 2016; Turolla et al., 2017) highlighting public and environmental risks posed by the acquirement of a functional genetic element by other microorganisms via HGT and hence other effective disinfection methods along with PAA (i.e., UV/PAA, UV-AOP, etc.) should be invented to ensure that not only host cell but also ARGs or plasmid get destroyed.

6.3. Ozonation

Ozone (O_3) is a powerful oxidizing agent that reacts with organic directly or indirectly via hydroxyl radicals ($HO^{\bullet}$) formation. A great deal of interest on ozonation showed the efficiency of ozone removing antibiotics, pharmaceutical compounds, ARBs, and ARGs (Espejo et al., 2014; Baghal Aaghari et al., 2021). Michael-Kordatou et al. (2017) observed rapid elimination of ethylparaben (EtP), erythromycin (ERY) and their resistant bacteria, where ERY degradation seemed to be driven by $HO^{\bullet}$ whereas EtP degradation was contributed by the O_3 and $HO^{\bullet}$ mediated pathways. Similarly, Iakovides et al. (2019) reported antibiotic degradation at a low dose of ozone (0.125 gO_3 $gDOC^{-1}$) and a significant inactivation of drug-resistant *E. coli* with simultaneous abundance reduction of examined ARGs (*aad*A1, *qac*ED1, *intl*1, *dfr*A1 and *sul*1) at higher ozone dose (0.25 gO_3 $gDOC^{-1}$) with hydraulic retention time (HRT) of 40 min. However, phytotoxicity analysis conducted by both the studies revealed augmentation of phytotoxicity events (i.e. root and shoot inhibition) governed by dissolved effluent organic matter (dE_fOM).

6.4. Photocatalytic processes

Photocatalytic processes are indirect processes where reactive oxygen species are generated ($HO_2^{\bullet}$, $HO^{\bullet}$, and $O_2^{\bullet-}$) that react with impurities such as organic, inorganic compounds, and biological entities (bacteria, viruses, etc.) leading to their decomposition (Etacheri et al., 2015; Sharma et al., 2016). TiO_2-based photocatalysts are widely used heterogeneous photocatalysts due to their non-toxicity, availability of various forms of TiO_2, photochemical stability as well as strong acid and base stability. Karaolia et al. (2018) synthesized TiO_2-reduced graphene oxide (TiO_2-rGO) composite photocatalysts by hydrothermal and photocatalytic treatment and showed that TiO_2-rGO-PH was more efficient in the photocatalytic degradation of clarithromycin (CLA) (86 ± 5%), and ERY (84 ± 2%) while degradation of SMX (87 ± 4%) was found to be slightly higher with pristine aeroxide P25 TiO_2 (artificial photocatalyst) under simulated solar radiation, in real urban wastewater effluents treated by a membrane bioreactor. In addition, complete inactivation was demonstrated with absence of post-treatment regrowth of *E. coli* (<LOD) even 24 h after the end of treatment with successful removal of *amp*C and reduced abundance of *ecf*X gene of *P. aeruginosa.* However, *erm*B, *sul*1, and 23 S rRNA for enterococci sequences were found to be persistent throughout the treatment. In another study, a series of photocatalyst based on TiO_2 (Au-TiO_2 and Pt-TiO_2) was tested for the elimination of *E. coli* from wastewater, where all the photocatalyst synthesized showed bactericidal activity with considerably better results using both bare and metalized TiO_2. Furthermore, total *E. coli* elimination and no bacterial regrowth was ere achieved (even after 72 h) with 3 h of photocatalytic reaction using 120 W m^2 of UV light intensity and 2 wt% Pt-TiO_2 as a photocatalyst, demonstrating an enhanced removal efficiency in presence of noble metals (Murcia et al., 2017).

6.5. Fenton processes

Photo-Fenton, solar photo-Fenton, and electro-Fenton are another promising approach that has proved to be effective in eliminating, and inactivating resistance-conferring plasmids (RCPs), micro-pollutants, antibiotics, ARGs, and ARBs including single drug-resistant and MDR bacteria (Giannakis et al., 2018; Ioannou-Ttofa et al., 2019; Ahmed et al., 2021; Vilela et al., 2021). Photo-Fenton is an AOP, based on $HO^{\bullet}$ generation from H_2O_2 catalyzed by iron salts (Fe^{2+}, Fe^{3+}) and UV–visible light. Subsequently, the process can be driven by solar radiation, hence called Solar Photo-Fenton (SPF) (Polo-López and Sánchez Pérez, 2021). Ahmed et al. (2020) reported a 6.17 log removal and no regrowth of *E. coli* DH5α harboring two plasmid-encoded ARGs (*tet*A and *bla*$_{TEM}$) applying photo-Fenton under visible LED and neutral pH conditions. Additionally, 6.75–8.56-log reduction in ARGs with degradation of *tet*A, *bla*$_{TEM}$ genes and shearing of extracellular DNA into fragments was observed. Similarly, Michael et al. (2019) reported complete antibiotics degradation, ARB inactivation, disabled bacterial regrowth, cellular DNA degradation, and decreased abundance of *bla*$_{CTX-M}$, *bla*$_{OXA}$, *qnr*S, *tet*M, and *sul*1 genes by solar photo-Fenton (SPF). However, SPF treatment resulted in increased toxicity toward *D. magna* probably due to the oxidation of dissolved effluent organic matter that led to the formation of toxic product. Therefore, SPF oxidation was followed by adsorption on granular activated carbon which provided almost complete removal of toxicity and antibiotics ensuring complete urban wastewater decontamination. Another study evaluated the efficacy of SPF in a raceway

Table 2
Treatment strategies to mitigate ARB and ARGs from the urban river ecosystem.

Treatment strategies	Dosage	Target bacteria	Target genes	Treatment efficiency	Main findings	References
H_2O_2/sunlight, TiO_2/sunlight, H_2O_2/TiO_2/sunlight, solar Photo-Fenton, and chlorination	Q_{UV} 8 kJ L^{-1}, Q_{UV} 3–5 kJ L^{-1} 1.0 mg Cl_2 L^{-1}, respectively	MDR *E. coli*	–	Total inactivation (100%)	Chlorination was found to be more effective compared to H_2O_2/sunlight in achieving total inactivation of MDR *E. coli*, but less effective in controlling their regrowth	(Fiorentino et al., 2015)
Solar stimulated N-doped TiO_2 photocatalysis	NDT photocatalyst at 0.2 g L^{-1} dose with 10 min of solar irradiation	Tetracycline, vancomycin, ciprofloxacin, and cefuroxime resistant *E. coli*	–	Inactivation rate 8.5×10^5 CFU 100 $mL^{-1}min^{-1}$ after 10 min	Decrease trend in resistance to ciprofloxacin and sensitivity to cefuroxime was observed after treatment. However, no significant effect of the solar photocatalytic process with NDT observed on tetracycline and vancomycin resistance of *E. coli*	(Rizzo et al., 2014)
Solar H_2O_2, photo-Fenton, TiO_2-25, and GO-TiO_2 mediated heterogeneous photocatalysis with or without H_2O_2	Quv < 30 kJL^{-1}	Total faecal coliforms, *E. coli*, enterococci, and their antibiotic-resistant (ciprofloxacin and tetracycline) counterparts	*qnrS*, bla_{CTX-M}, bla_{TEM}, 16 S rRNA, *IntI1, sul1*	Reduction rate (1.0 CFU mL^{-1})	P25/H_2O_2 and solar-H_2O_2 were found to be most efficient in the reduction of abundance (gene copy number per volume of wastewater) of analyzed genes. Nevertheless, this reduction was transient for *sul1,* IntI1*, and* 16 S rRNA genes, since an abundance of these genes were increased to values close to pre-treatment after 3 days of storage. A similar pattern was also observed for, bla_{CTX-M}, bla_{TEM} using TiO_2-P25, TiO_2-P25/H_2O_2, and *qnrS* using TiO_2-P25.	(Moreira et al., 2018)
Photo-Fenton like process using cu-iminodisuccinic acid complex (UV-C/H_2O_2/Cu-IDS), and UV-C/H_2O_2/Fe, UV-C/H_2O_2/Cu, H_2O_2, and UV-C	Cu-IDS complex (0.01 and 0.02 mM), copper sulphate (as source of Cu^{2+}, 0.008 and 0.016 mM) and ferrous sulphate (as source of Fe^{2+}, 0.018 and 0.035 mM)	*E. coli*	–	Inactivation rate (3.5 log units in 10 min at neutral pH 7.8 ± 0.5)	UV-C/H_2O_2/Cu-IDS is effective under the real condition in urban wastewater thus overcoming the main obstacle of acidic pH to use the Photo-Fenton process.	(Fiorentino et al., 2018)
TiO_2-photocatalytic treatment using UVA-LEDs	4 LEDs symmetrically distributed and 1.00 gL^{-1} of catalyst	Total and resistant heterotrophs, *E. coli,* and enterococci	–	Decreased bacterial load by 2 log-units after 1 hr treatment	Although this apparent removal was not enough to prevent Bacterial regrowth of total heterotrophs to value close to pre-treatment. However, the percentage of antibiotic resistance after regrowth was similar (for *E. coli, and* total heterotrophs) or lower than untreated urban wastewater suggesting a potential of process for wastewater treatment but, conditions need to be adjusted to minimize bacterial regrowth.	(Biancullo et al., 2019)
Fenton oxidation ($Fe^{2+/}H_2O_2$) and UV/H_2O_2 process	$Fe^{2+/}H_2O_2$ a molar ratio of 0.1, H_2O_2 concentration of 0.01 mol L^{-1}, pH of 3.0, a reaction time of 2 h	–	*sul1*, *tet*X, *tet*G, IntI1, and 16 S rRNA	2.58–3.79 log bacterial reduction for $Fe^{2+/}H_2O_2$ and 2.8–3.5 log reduction for UV/H_2O_2	Both Fenton oxidation ($Fe^{2+/}H_2O_2$) and UV/H_2O_2 process are effective in selectively reducing ARGs. Removal of target genes affected by many factors reaction time, $Fe^{2+/}H_2O_2$ molar ratios, solution pH, and H_2O_2 concentration, among which reagent concentration and pH are most important	(Zhang et al., 2016)
Two vertical subsurface flow (VF) constructed wetlands [an unsaturated unit (UVF) and a partially saturated unit (SVF) operating in parallel]	organic loading rate (OLR) and hydraulic loading rate (HLR) of 30 g COD m− 2 day^{-1} and 133 mm day^{-1}	–	bla_{TEM}, IntI1, *sul1*, *sul2*, *qnr*S, *erm*B, and 16 S rRNA	93% removal rate	Significantly higher removal of fluoroquinolones, metronidazole, and azithromycin antibiotics observed in SVF compared to UVF. While removal efficiency % of the sulphonamides, trimethoprim, clarithromycin, ampicillin was similar in both units or slightly higher in UVF.	(Ávila et al., 2021)

pond reactor at neutral pH and detected efficient removal antibiotics (60–100%) after 3 h with total bacterial inactivation in the range 30–100 min (3.2–6.7 KJ L^{-1}). However, SPF was found to be poorly effective in the removal of ARGs (sulfonamide resistance genes reduced to some extent (relative abundance <1), and no effect on class 1 integron *intI1* and ß-lactam resistance genes) suggesting a need for more intensive oxidative conditions for efficient removal of ARGs (Fiorentino et al., 2019).

6.6. Constructed wetlands

Constructed wetlands are engineered aquatic systems with diverse microbial communities and are used to treat wastewater by the same biogeochemical processes dominant in natural wetlands (Uluseker et al., 2021) and can be classified into three main types depending on the pattern of wastewater flowing through the constructed wetlands: free surface flow, vertical subsurface flow, and horizontal subsurface flow (Sharma et al., 2016). Zhang et al. (2020) investigated removal of eight antibiotics from urban wastewater and reported efficient removal of antibiotics with their median in following order ofloxacin (92%) > lomefloxacin (86%) > tetracycline (71%) > erythromycin (65%) > roxithromycin (61%) > oxytetracycline (59%) > norfloxacin (56%) > ciprofloxacin (32%). Similar results were obtained in a study by Hijosa-Valsero et al. (2011) reporting removal of doxycycline (61 ± 38%), sulfamethoxazole (60 ± 26%) and amoxicillin (45 ± 15%) from urban wastewater. Correspondingly, multiple studies have documented the effective elimination of ARGs by the use of constructed wetlands (Chen et al., 2016; Abou-Kandil et al., 2021; Sabri et al., 2021). High accumulation of antibiotics and long-term stress on constructed wetland provide a favorable setting for the emergence and spread of ARGs, which may result in regrowth of resistance carrying bacteria (Liu et al., 2019a).

Sun et al. (2019) and Liu et al. (2021) reviewed different treatment technologies applied in WWTP to remove microplastics, that includes primary (primary settling treatment, grit and grease treatment), secondary (A^2 O, biofilters, and other bioreactors) and tertiary treatment strategies (UV, O_3, chlorination, biologically active filters, disc filters, and rapid sand filters).

Even though some significant progress has been made in the effective treatment of wastewater, major key points should be taken into account to improve the efficiency of anthropogenic waste, ARGs, and ARB removal. Combination strategies (such as UV/PAA, photocatalytic ozonation, PAA/AOP, UV/chlorine) may show better performance in wastewater treatment. However, investigations are needed for deeper understanding and full potential exploration of combinatorial strategies. However, some new studies have emerged recently, for instance Wang et al. (2020b) effectively overcome a major drawback of UV disinfection, the photoreactivation using combination of UV/chlorine resulting in reduction of plasmid conjugation transfer and removal of ARGs. Thus, development of a new or combined strategy and further optimization might be another overcome for future research work to significantly remove/degrade anthropogenic waste and simultaneously to control the emergence of AMR in the environment. Need for extensive research focusing on functionality and degradation of ARGs, plasmids, and MGEs after treatment is nowadays a prime research mainly focused. Identifying and understanding the effect of diverse chemical, biological and microbiological derived components of wastewater on the efficiency of treatment strategies such as decreased PAA inactivation of MDR bacteria by soluble constituents of wastewater was investigated by Balachandran et al. (2021). Microplastics are emerging as a major driver of AMR and ARGs in river ecosystems and their removal may be achieved through the use of strategies like microbial degradation, use of natural products coupled with in silico approaches and nanomaterials (Kaur et al., 2021). Difficulty in scale-up due to use of expensive chemicals, complex protocol preparation of nanostructures/nanocomposite (e.g., TiO_2-based nanocomposite) utilized during treatment and high energy requirement involved in supplying visible and UV light for the treatment with probable threat of selection posed by application metals (TiO_2, Fe, Au, Pt, etc.) makes the mitigation more difficult (Michael-Kordatou et al., 2018; Wang et al., 2020b).

7. Conclusion

Given the present condition of urban river pollution and its wide-ranging environmental ramifications, more research is needed into the issues, particularly in terms of reducing their environmental impacts. In terms of antimicrobial resistance, urban river ecosystem receives less attention than other water bodies. However, the finding of the limited research show unquestionable role of contaminants in the emergence, prevalence and spread of antibiotics, ARB and ARGs in the urban river. ARGs is a natural phenomenon, and humans have been living with them in the environment (and within our microbiome) since the start of our species. Apart from factors such as antibiotics, disinfection by-products (DBPs), heavy metal genes, and microplastics, environmental factors such as sub-MIC selection, co-selection, co-occurrence, sub-lethal effect, and low levels/ dissemination of antibiotics are also considered as the major AMR driver pathways. Urban River being the significant reservoirs of ARGs are important for disseminating them with MGEs playing an important role in the AMR development and emergence. Even though significant progresses and measures such as peracetic acid disinfection, constructed wetlands, photocatalytic processes, ozonation and Fenton processes have been introduced in the effective treatment of urban rivers, newer technologies are demanded to mitigate AMR and the current threat of antimicrobial resistance to human health.

CRediT authorship contribution statement

All the authors designed, edit review and revise the review.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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